

Cauchy and Sylow's Theorems

Vincent Wang

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1 Problems

Cauchy's theorem. Let G be a finite group with $p \mid |G|$. Then there exists an element $g \in G$ such that $|g| = p$.

Proof. We proceed by induction on $|G|$. For the base case, G is cyclic of order p . Assume $p \mid |G|$ and the result for all groups smaller than G . We case on $Z(G)$.

First, suppose $p \mid |Z(G)|$. By the FTFGAG, there exists a cyclic subgroup $C_p^k < Z(G)$. If $C_{p^k} = \langle y \rangle$, then $|y^{p^{k-1}}| = p$.

Now suppose $p \nmid |Z(G)|$ and $Z(G) \neq e$. Then consider the group $G/Z(G)$. We may apply the IH because $|G/Z(G)| = |G|/|Z(G)| < |G|$ and p divides the order of this quotient group. By the IH, there exists $g \in G$ such that $g^k \notin Z(G)$ for $k < p$ and $g^p \in Z(G)$. Consider the g -conjugation action on G . By the Class equation, $|Z(g)| + \sum_{c \in \mathcal{C}} |c| = |G|$. Each (nonsingular) conjugacy class has order p and $p \mid |G|$, so $p \mid |Z(g)|$. Note $Z(g) \subsetneq G$ because that would mean $gZ(G) = Z(G)$. Hence $Z(g)$ is a proper subgroup of G , for which we apply the IH again and resolve this case.

Suppose $Z(G) = e$. Consider any nontrivial element h of G , say with order n . Pick a prime divisor q of n , so that $g = h^{n/q}$ has order q . If $q = p$, we're done, so assume otherwise. Consider the g conjugation action on G . By the Class equation, $|Z(g)| + \sum_{c \in \mathcal{C}} |c| = |G|$. Since $|Z(g)| < |G|$ as $Z(G) = e$, we have some conjugacy class c . Take any $x \in c$ and consider the subgroup $\text{Stab}(x) < G$. The orbit of x is c , which is size q . By Orbit-stabilizer theorem, $|\text{Stab}(x)| = |G|/q$, so we apply the IH to this stabilizer group, and we're done.

Second magical proof. Consider the action of $C_p = \langle \varphi \rangle$ on the set of p -tuples $(g_1, g_2, \dots, g_p) \in G^p$ such that $g_1 g_2 \cdots g_p = e$. The elements fixed by φ are elements $S = \{g \in G : g^p = e\}$. The number of orbits of the remaining elements is

$$\frac{|G|^{p-1} - |S|}{p}.$$

Since $p \mid |G|$, we have $p \mid |S|$, for which there are at least $p - 1$ nontrivial elements in S , and we're done.

Generalizations. Consider the action of C_N , where $N = p^2$, on the set of N -tuples $(g_1, g_2, \dots, g_N)$ for which $g_1 g_2 \cdots g_N = e$. By Burnside's Lemma, the number of Orbits is

$$\mathcal{O} = \frac{1}{p^2} (|G|^{N-1} + (p-1)|G|^{p-1}|S_1| + (p^2-p)|S_2|),$$

where S_1, S_2 denote the sets of elements for which $g^p = e$ and $g^{p^2} = e$, respectively. Our work above has shown $p \mid |S_1|$, so $(p-1)|G|^{p-1}|S_1|$ is divisible by p^p or p^2 . $N = p^2 - 1 \geq 3$, so $|G|^{N-1}$ is divisible by p^2 . Hence, $p^2 \mid p|S_2|$, or $p \mid |S_2|$. This generalizes.

Let N_n denote the number of elements g in G for which $g^{p^n} = e$. I claim that $p \mid N_n$ for all positive n . We proceed by induction. The base case is shown above.

Let $M = p^{n+1}$. Consider the $C_M = \langle \varphi \rangle$ action on the set X of M -tuples $(x_1, x_2, \dots, x_M)$ such that $x_1 x_2 \cdots x_M = e$. We will apply Burnside's Lemma to count the number of C_M -orbits on X .

(Fill in details for Burnside's Lemma later) By Burnside's Lemma, the number of orbits is

$$|\mathcal{O}| = \frac{1}{p^{n+1}} \left(|G|^{p^{n+1}-1} + \left(\sum_{i=1}^n |G|^{p^{n+1-i}-1} (p^i - p^{i-1}) N_i \right) + (p^{n+1} - p^n) N_{n+1} \right).$$

Each of the middle terms cancel because $p^{n+1-i} \geq n+1-i$ and $(p^i - p^{i-1}) N_i$ each contribute a factor of p^i by the IH. The first term is divisible by p^{n+1} because $p^{n+1} - 1 \geq n+1$. Hence, p^{n+1} divides the last term, for which $p^{n+1} \mid p^n N_{n+1}$, or $p \mid N_{n+1}$, completing the induction.